

13.1 Math and Art



Figure 13.2 Sunflower seeds appear in a pattern that involves the Fibonacci sequence. (credit: “Sunflower Surprise” by frankieleon/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify and describe the golden ratio.
2. Identify and describe the Fibonacci sequence and its application to nature.
3. Apply the golden ratio and the Fibonacci sequence relationship.
4. Identify and compute golden rectangles.

Art is the expression or application of human creative skill and imagination, typically in a visual form such as painting or sculpture, producing works to be appreciated primarily for their beauty or emotional power.

Oxford Dictionary

Art, like other disciplines, is an area that combines talent and experience with education. While not everyone considers themselves skilled at creating art, there are mathematical relationships commonly found in artistic masterpieces that drive what is considered attractive to the eye. Nature is full of examples of these mathematical relationships.

Enroll in a cake decorating class and, when you learn how to create flowers out of icing, you will likely be directed as to the number of petals to use. Depending on the desired size of a rose flower, the recommendation for the number of petals to use is commonly 5, 8, or 13 petals. If learning to draw portraits, you may be surprised to learn that eyes are approximately halfway between the top of a person's head and their chin. Studying architecture, we find examples of buildings that contain golden rectangles and ratios that add to the beautifying of the design. The Parthenon ([Figure 13.3](#)), which was built around 400 BC, as well as modern-day structures such as the Washington Monument are two examples containing these relationships. These seemingly unrelated examples and many more highlight mathematical relationships that we associate with beauty in artistic form.



Figure 13.3 The Parthenon in Greece demonstrates the golden ratio. (credit: "Parthénon" by Julien Maury/Flickr, Public Domain Mark 1.0)

Golden Ratio

The **golden ratio**, also known as the golden proportion, is a ratio aspect that can be found in beauty from nature to human anatomy as well as in golden rectangles that are commonly found in building structures. The golden ratio is expressed in nature from plants to creatures such as the starfish, honeybees, seashells, and more. It is commonly noted by the Greek letter ϕ (pronounced "fee"). $\phi = \frac{1+\sqrt{5}}{2}$, which has a decimal value approximately equal to 1.618.

Consider [Figure 13.3](#): Note how the building is balanced in dimension and has a natural shape. The overall structure does not appear as if it is too wide or too tall in comparison to the other dimensions.

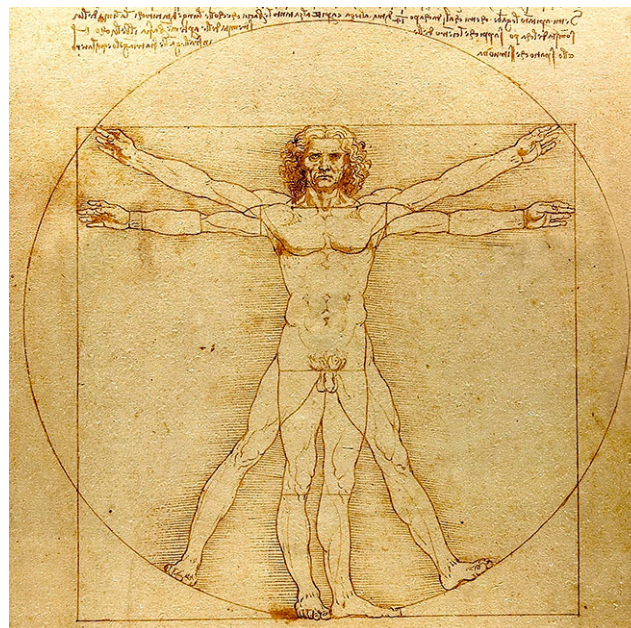


Figure 13.4 *Vitruvian Man* by Leonardo da Vinci (credit: "Vitruvian Man" by Leonardo da Vinci/Wikimedia Commons, Public Domain)

The golden ratio has been used by artists through the years and can be found in art dating back to 3000 BC. Leonardo da Vinci is considered one of the artists who mastered the mathematics of the golden ratio, which is prevalent in his artwork such as *Vitruvian Man* ([Figure 13.4](#)). This famous masterpiece highlights the golden ratio in the proportions of an ideal body shape.

The golden ratio is approximated in several physical measurements of the human body and parts exhibiting the golden ratio are simply called golden. The ratio of a person's height to the length from their belly button to the floor is ϕ or approximately 1.618. The bones in our fingers (excluding the thumb), are golden as they form a ratio that approximates ϕ . The human face also includes several ratios and those faces that are considered attractive commonly exhibit golden ratios.

EXAMPLE 13.1**Using Golden Ratio and a Person's Height**

If a person's height is 5 ft 6 in, what is the approximate length from their belly button to the floor rounded to the nearest inch, assuming the ratio is golden?

✓ **Solution**

Step 1: Convert the height to inches

$$5 \text{ ft } 6 \text{ in} = 66 \text{ in}$$

Step 2: Calculate the length from the belly button to the floor, L .

$$66/L = 1.618$$

$$L = 40.8 \text{ in}$$

The length from the person's belly button to the floor would be approximately 41 in.

**YOUR TURN 13.1**

1. If a person's height is 6 ft 2 in, what is the approximate length from their belly button to the floor rounded to the nearest inch if the ratio is golden?

Fibonacci Sequence and Application to Nature

Figure 13.5 Rose petals appear in a Fibonacci spiral. (credit: "rilke4" by monchoohcnom/Flickr, Public Domain Mark 1.0)

The **Fibonacci sequence** can be found occurring naturally in a wide array of elements in our environment from the number of petals on a rose flower to the spirals on a pine cone to the spines on a head of lettuce and more. The Fibonacci sequence can be found in artistic renderings of nature to develop aesthetically pleasing and realistic artistic creations such as in sculptures, paintings, landscape, building design, and more. It is the sequence of numbers beginning with 1, 1, and each subsequent term is the sum of the previous two terms in the sequence (1, 1, 2, 3, 5, 8, 13, ...).

The petal counts on some flowers are represented in the Fibonacci sequence. A daisy is sometimes associated with plucking petals to answer the question "They love me, they love me not." Interestingly, a daisy found growing wild typically contains 13, 21, or 34 petals and it is noted that these numbers are part of the Fibonacci sequence. The number

of petals aligns with the spirals in the flower family.

EXAMPLE 13.2

Applying the Fibonacci Sequence to Rose Petals

Suppose you were creating a rose out of icing, assuming a Fibonacci sequence in the petals, how many petals would be in the row following a row containing 13 petals?

✓ Solution

The number of petals on a rose is often modeled with the numbers in the Fibonacci sequence, which is 1, 1, 2, 3, 5, 8, 13, ..., where the next number in the sequence is the sum of $8 + 13 = 21$. There would be 21 petals on the next row of the icing rose.

> YOUR TURN 13.2

1. If a circular row on a pinecone contains 21 scales and models the Fibonacci sequence, approximately how many scales would be found on the next circular row?

Golden Ratio and the Fibonacci Sequence Relationship

Mathematicians for years have explored patterns and applications to the world around us and continue to do so today. One such pattern can be found in ratios of two adjacent terms of the Fibonacci sequence.

Recall that the Fibonacci sequence = 1, 1, 3, 5, 8, 13, ... with 5 and 8 being one example of adjacent terms. When computing the ratio of the larger number to the preceding number such as $8/5$ or $13/8$, it is fascinating to find the golden ratio emerge. As larger numbers from the Fibonacci sequence are utilized in the ratio, the value more closely approaches ϕ , the golden ratio.

EXAMPLE 13.3

Finding Golden Ratio in Adjacent Fibonacci Terms

The 24th Fibonacci number is 46,368 and the 25th is 75,025. Show that the ratio of the 25th and 24th Fibonacci numbers is approximately ϕ . Round your answer to the nearest thousandth.

✓ Solution

$75,025/46,368 = 1.618$; The ratio of the 25th and 24th term is approximately equal to the value of ϕ rounded to the nearest thousandth, 1.618.

> YOUR TURN 13.3

1. The 23rd Fibonacci number is 28,657 and the 24th is 46,368. Show that the ratio of the 24th and 23rd Fibonacci numbers is approximately ϕ . Round your answer to the nearest thousandth.



Figure 13.6 The pyramids of Giza in Egypt (credit: “Giza Pyramids” by Vincent Brown/Flickr, CC BY 2.0)

Golden Rectangles

Turning our attention to man-made elements, the golden ratio can be found in architecture and artwork dating back to the ancient pyramids in Egypt (Figure 13.6) to modern-day buildings such as the UN headquarters. The ancient Greeks used **golden rectangles**—any rectangles where the ratio of the length to the width is the golden ratio—to create aesthetically pleasing as well as solid structures, with examples of the golden rectangle often being used multiple times in the same building such as the Parthenon, which is shown in Figure 13.3. Golden rectangles can be found in twentieth-century buildings as well, such as the Washington Monument.

Looking at another man-made element, artists' paintings often contain golden rectangles. Well-known paintings such as Leonardo da Vinci's *The Last Supper* and the *Vitruvian Man* contain multiple golden rectangles as do many of da Vinci's masterpieces.

Whether framing a painting or designing a building, the golden rectangle has been widely utilized by artists and are considered to be the most visually pleasing rectangles.

EXAMPLE 13.4

Finding Golden Rectangle in Frames

A frame has dimensions of 8 in by 6 in. Calculate the ratio of the sides rounded to the nearest thousandth and determine if the size approximates a golden rectangle.

Solution

$8/6 = 1.333$; A golden rectangle's ratio is approximately 1.618. The frame dimensions are close to a golden rectangle.



YOUR TURN 13.4

1. A frame has dimensions of 10 in by 8 in. Calculate the ratio of the sides rounded to the nearest thousandth and determine if the size approximates a golden rectangle.



PEOPLE IN MATHEMATICS

M.C. Escher

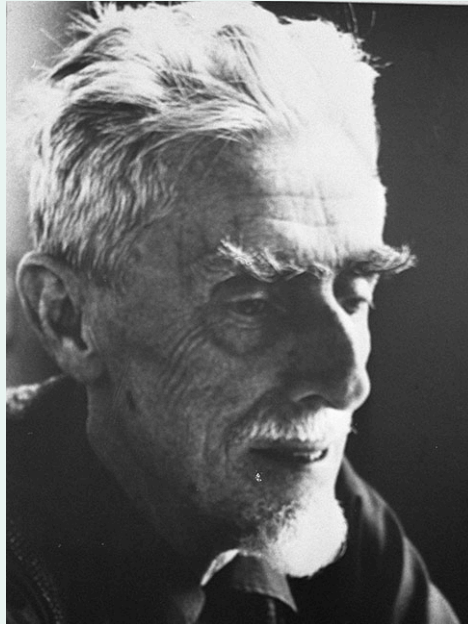


Figure 13.7 M.C. Escher (credit: "M.C. Escher" by Hans Peters (ANEFO)/Dutch National Archives, CC0 1.0 Public Domain)

Maurits Cornelis Escher was a Dutch-born world-famous graphic artist and his work can be found in murals, stamps, wallpaper designs, illustrations in books, and even carpets. Over his lifetime, M.C. Escher created hundreds of lithographs and wood engravings as well as more than 2,000 sketches.

Escher's work is characterized with the infusion of geometric designs that obey most of the mathematical rules. If you study his work closely, you can see where he breaks a mathematical relationship to create famous illusions such as soldiers marching around the top of a square turret where the soldiers appear to be always going uphill but are contained on a single set of stairs in a square. Look closely and the golden ratio as well as golden rectangles abound in Escher's work.

Like many famous people, M.C. Escher did not find success in his early school years. Before finding success, Escher failed his final school exam and quit a short stint in architecture. Finding a graphic arts teacher who recognized Escher's talent, Escher completed art school and enjoyed traveling through Italy, where he found much of his inspiration for his work.

Check Your Understanding

1. What is the value of the golden ratio to the nearest thousandth?
2. What are the first 10 terms of the Fibonacci sequence?
3. What is a golden rectangle?



SECTION 13.1 EXERCISES

1. A person's height is 5 ft 2 in. What is the approximate length from their belly button to the floor rounded to the nearest inch?
2. A person's height is 6 ft 3 in. What is the approximate length from their belly button to the floor rounded to the nearest inch?
3. A person's length from their belly button to the floor is 3 ft 11 in. What is the person's approximate height rounded

to the nearest inch?

4. A person's length from their belly button to the floor is 58 in. What is the person's approximate height rounded to the nearest inch?
5. The spikes on a pineapple mirror the Fibonacci sequence. If a row on a pineapple contains five spikes, approximately how many spikes would be found on the next larger row of spikes?
6. The leaves on a plant mirror the Fibonacci sequence. If a set of leaves on the plant contains 5 leaves, how many leaves would be found on the previous smaller set of leaves?
7. The spines on a head of lettuce mirror the Fibonacci sequence. If a head of lettuce contains 13 spines, approximately how many spines would be found on the next inside layer?
8. The seeds on a sunflower mirror the Fibonacci sequence. If a circular layer on the sunflower contains 55 seeds, approximately how many seeds would be found on the next larger circular layer?
9. The segments on a palm frond mirror the Fibonacci sequence. If a palm frond contains 89 segments, approximately how many segments would be found on the next larger palm frond?
10. The 19th term of the Fibonacci sequence is 4,181 and the 20th term is 6,765. What is the 21st term of the sequence?
11. The 23rd term of the Fibonacci sequence is 28,657 and the 24th term is 46,368. What is the 22nd term of the sequence?
12. The 18th term of the Fibonacci sequence is 2,584 and the 20th term is 6,765. What is the 19th term of the sequence?
13. The 25th term of the Fibonacci sequence is 75,025 and the 20th term is 6,765. What is the 24th term of the sequence?
14. The 10th Fibonacci number is 55 and the 11th is 89. Show that the ratio of the 11th and 10th Fibonacci numbers is approximately ϕ . Round your answer to the nearest thousandth.
15. The 23rd Fibonacci number is 28,657 and the 24th is 46,368. Show that the ratio of the 24th and 23rd Fibonacci numbers is approximately ϕ . Round your answer to the nearest ten-thousandth.
16. The 22nd Fibonacci number is 17,711 and the 21st is 10,946. Show that the ratio of the 22nd and 21st Fibonacci numbers is approximately ϕ . Round your answer to the nearest ten-thousandth.
17. The 16th term of the Fibonacci sequence is 987. Use the approximate value of ϕ of 1.618 to estimate the 15th term. Round your answer to the nearest whole number.
18. The 26th term of the Fibonacci sequence is 121,393. Use the approximate value of ϕ of 1.618 to estimate the 25th term. Round your answer to the nearest whole number.
19. A frame has dimensions of 20 in by 24 in. Calculate the ratio of the sides rounded to the nearest tenth and determine if the size approximates a golden rectangle.
20. A fence has dimensions of 75 in by 45 in. Calculate the ratio of the sides rounded to the nearest tenth and determine if the size approximates a golden rectangle.
21. A frame has a length of 50 in. Calculate the width rounded to the nearest inch if the frame is to be a golden rectangle.